"Pawsitive Connect"

System Documentation

Developers

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1. Introduction

Pawsitive Connect is an enterprise-grade, full-stack native desktop application engineered to digitize and optimize animal rescue, shelter management, veterinary welfare, and pet adoption workflows. Designed as a unified ecosystem, the platform bridges the gap between stray/abandoned animal rescue operations, medical health tracking, shelter inventory management, and prospective adopters. By coupling a high-performance JavaFX desktop frontend with a decoupled Node.js REST backend and a spatial PostgreSQL database, Pawsitive Connect ensures reliable and secure animal welfare coordination.

Pawsitive Connect successfully unifies animal rescue administration into a modular, decoupled architecture. By separating the frontend presentation from backend microservices — the system guarantees high maintainability, strict boundary rules, and robust audit capabilities.

2. Existing Problem

Traditional animal rescue organizations and shelters frequently struggle with fragmented operational workflows, leading to inefficiencies in animal care and placement. Key challenges include:

- **Decentralized Data Management:** Reliance on manual paper logs, unorganized spreadsheets, and isolated notes makes tracking animal health, vaccination schedules, and adoption statuses error-prone.
- **Coordination Delays in Emergencies:** Distress calls regarding stranded or injured animals lack real-time geographic tracking and structured dispatch protocols, delaying critical rescue response times.
- **Opaque Adoption & Welfare Workflows:** Prospective adopters experience prolonged waiting periods and unclear feedback loops due to disconnected identity verification, background checks, and welfare screening steps.
- **Financial & Data Inconsistency:** Lacking strict database boundaries and atomic transaction ledgers, shelters face risks of record corruption and untracked financial transactions.

3. Software Requirement Specification (SRS)

Functional Requirements

1. **User Identity & Authentication:** Secure user registration, authentication via JSON Web Tokens (JWT), password hashing using `bcryptjs`, and role-based permissions.
2. **Pet Catalogue & Inventory:** Dynamic browsing, pagination, and multi-parameter filtering (by breed, age, status, and shelter branch) of rescue animals.
3. **Welfare & Adoptions Management:** Comprehensive logging of medical history, vaccinations, fundraising campaigns, and end-to-end processing of adoption applications.
4. **Emergency Rescue & GPS Dispatch:** Real-time logging of distress incidents, coordinate validation via PostGIS spatial extensions, and status transitions from `REPORTED` to `DISPATCHED`.
5. **Transactions & Financial Ledger:** Secure tracking and settlement of adoption fees, donations, and financial audit trails.

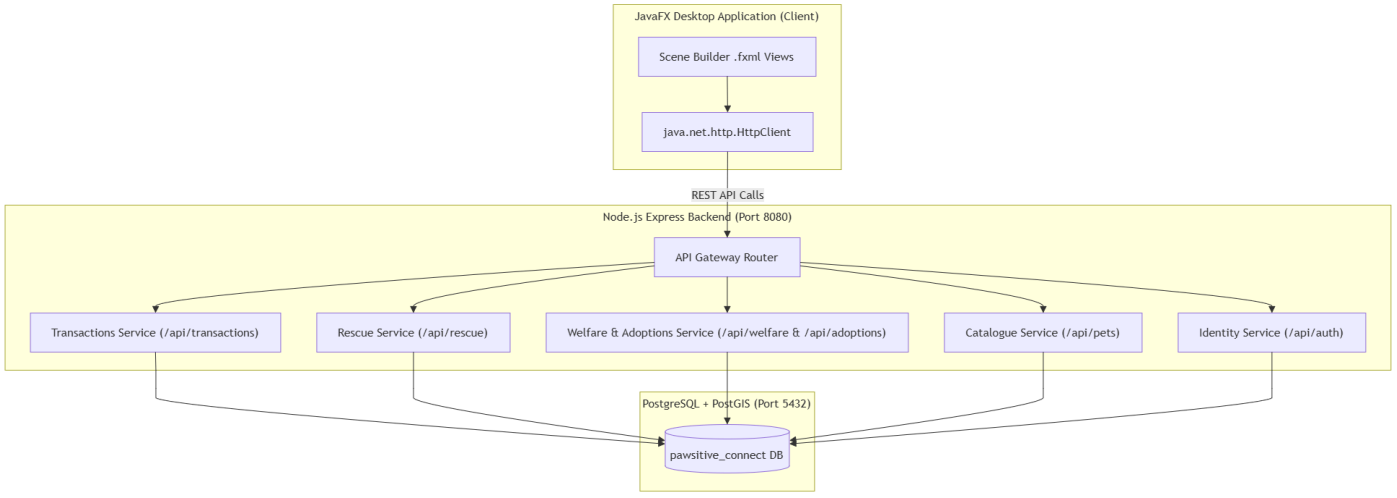
Non-Functional Requirements

- **Performance & Responsiveness:** Instant UI feedback on JavaFX threads using asynchronous HTTP clients (`java.net.http.HttpClient`) to prevent interface freezing during network calls.
- **Security & Reliability:** Strict payload validation, standardized RFC 7807 Problem Details error responses, and exclusion of sensitive database internals from client-facing messages.
- **Portability:** Packaged as a standalone native Windows executable (`PawsitiveConnect_Setup.exe`) using JDK `jpackage` and the WiX Toolset.

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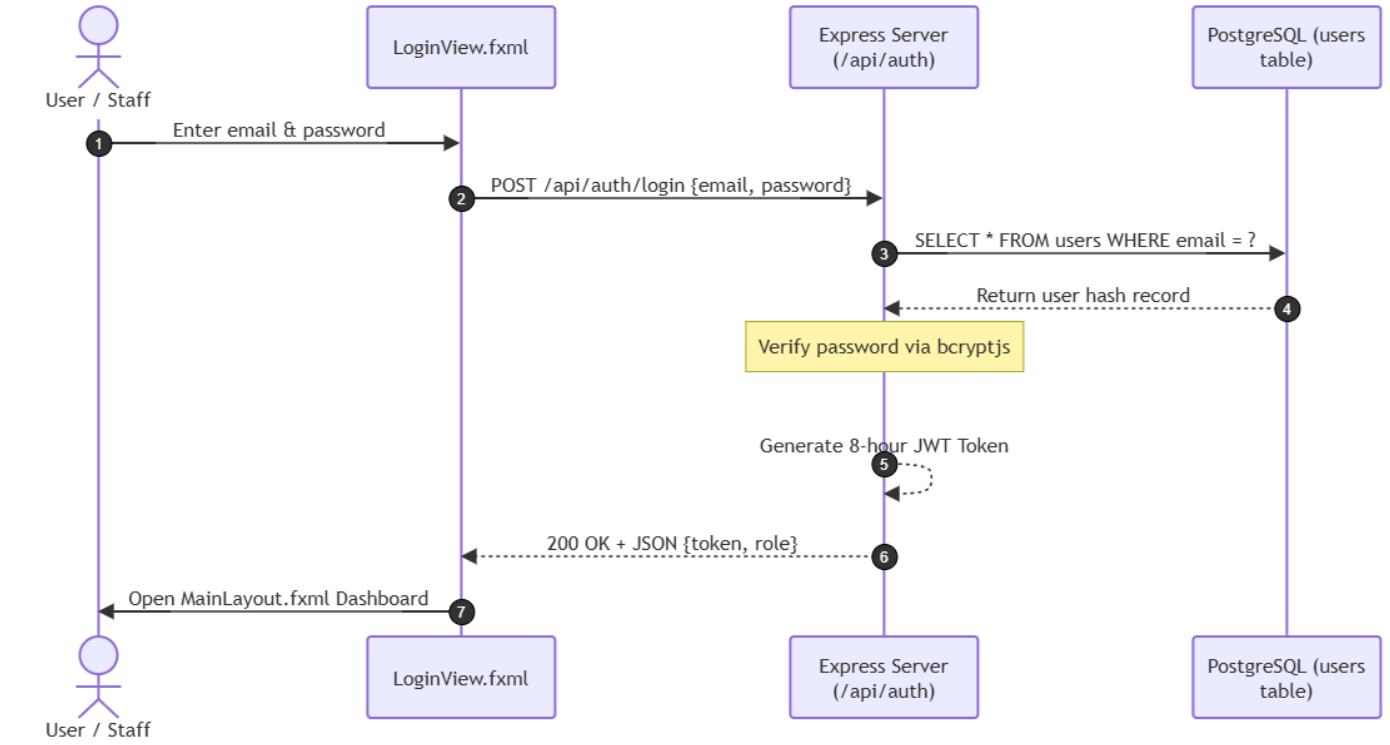
4. UML Diagram

Class Architecture Diagram

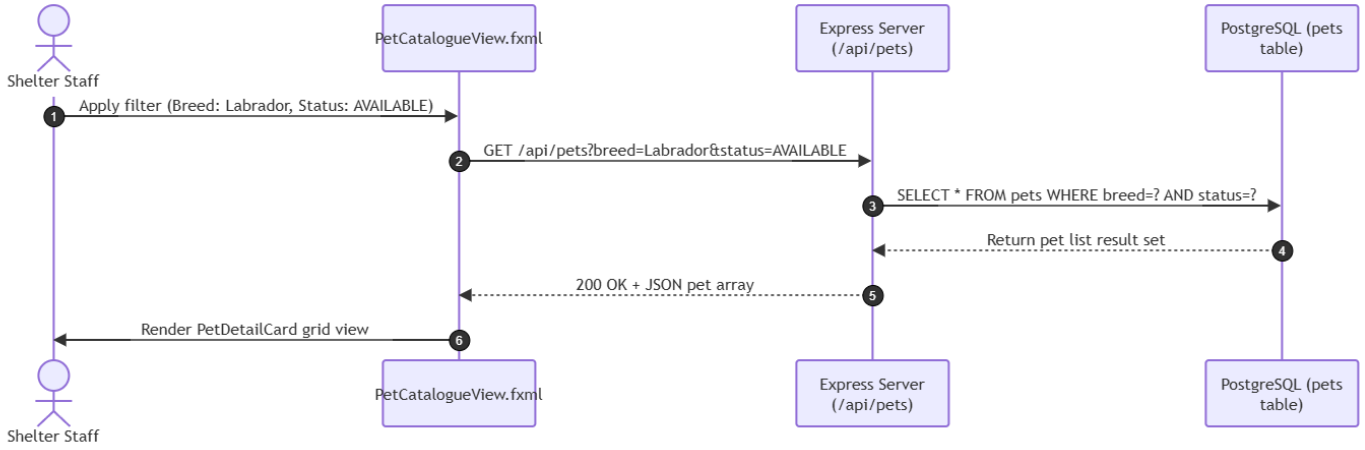


Sequence Diagrams for All Services

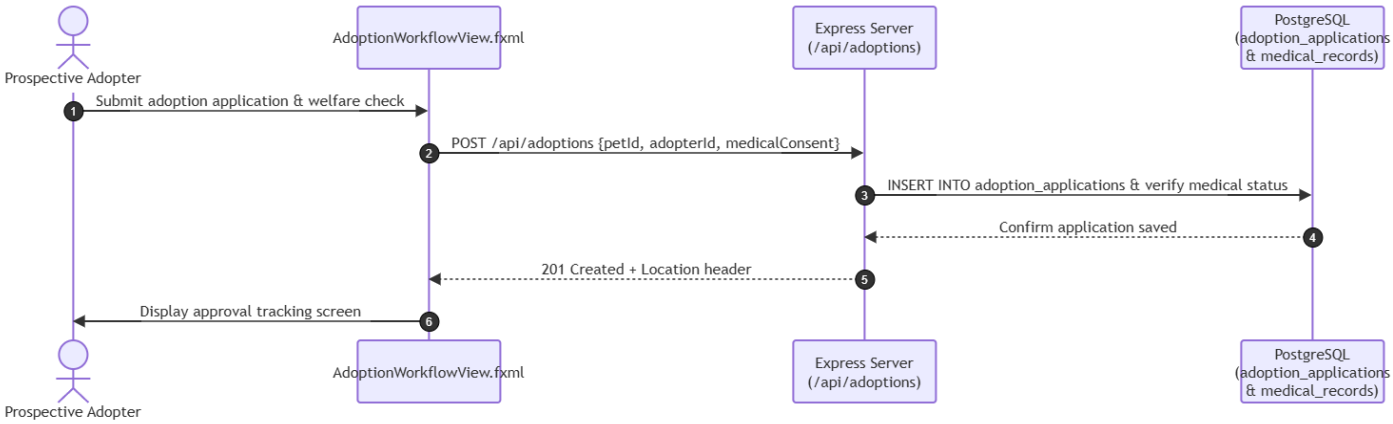
- **Identity Service** (`/api/auth/login`)



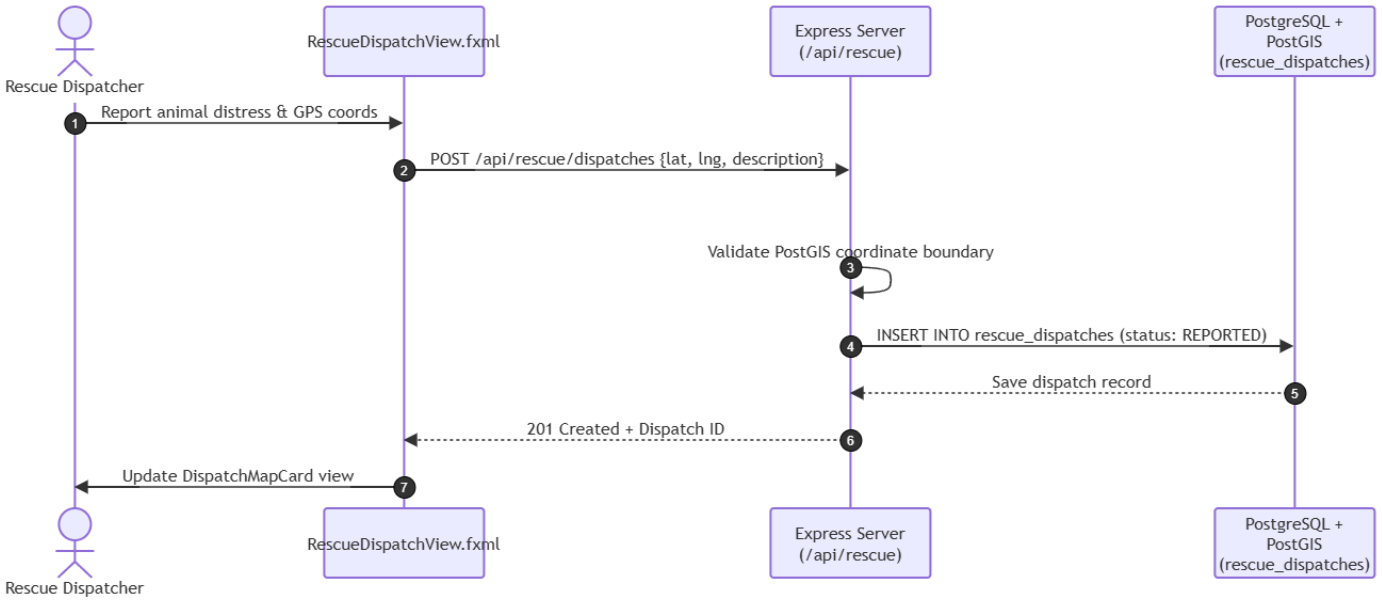
- **Catalogue Service** (`/api/pets`)



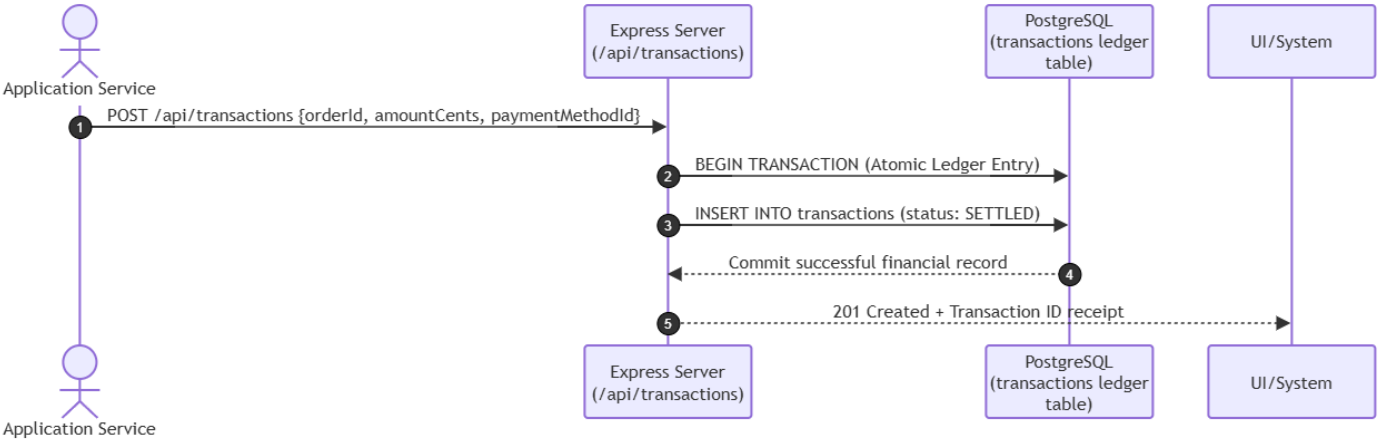
- **Welfare & Adoptions Service** (`/api/adoptions` & `/api/medical`)



- **Rescue Service** (`/api/rescue`)



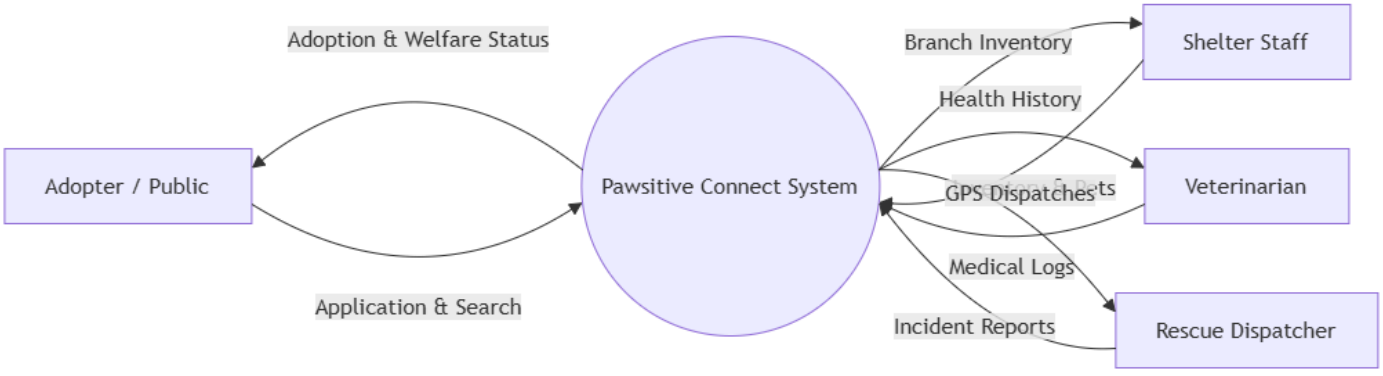
- **Transactions Service** (`/api/transactions`)



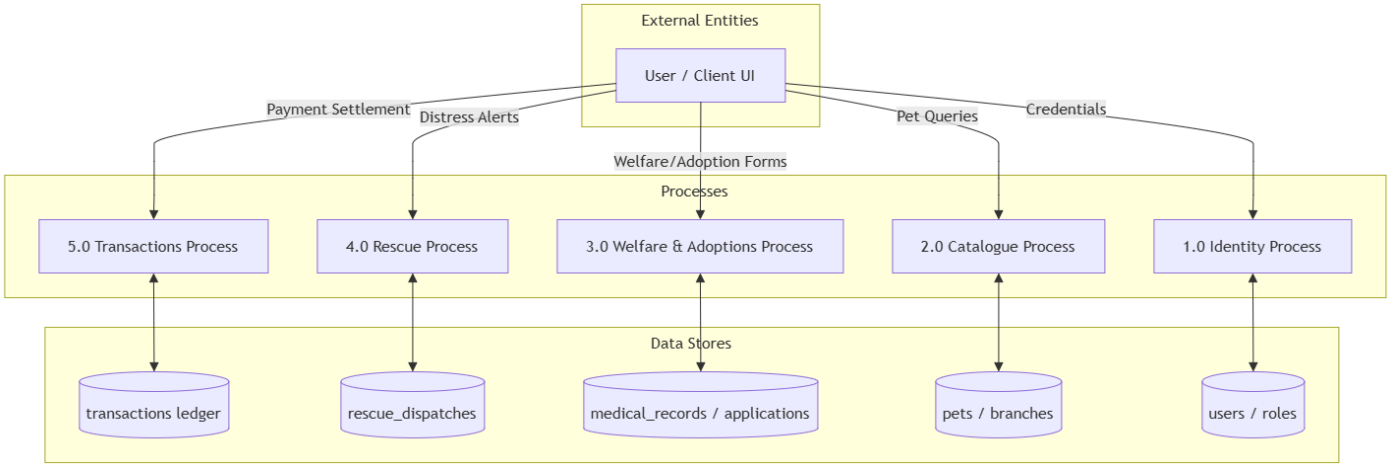
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5. Data Flow Diagram (DFD)

Level 0 Data Flow Diagram (Context Diagram)



Level 1 Data Flow Diagram (Service Breakdown Diagram)



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6. Technology Stack

- **Frontend Layer:** JavaFX (Java 21 LTS), FXML layouts via Gluon Scene Builder, and Maven build automation.
- **Backend Layer:** Node.js & Express.js REST API server running on **Port 8080**, utilizing JSON Web Tokens and `bcryptjs`.
- **Database Layer:** PostgreSQL relational database running on **Port 5432**, enhanced with the PostGIS spatial extension.
- **Packaging Toolset:** JDK `jpackage` and WiX Toolset for generating `PawsitiveConnect_Setup.exe`.

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7. Conclusion

Pawsitive Connect designed as a unified ecosystem, the platform bridges the gap between stray/abandoned animal rescue operations, medical health tracking, shelter inventory management, and prospective adopters. It provides a complete, scalable technological solution to modern animal welfare and rescue challenges. Through rigorous domain modeling, clear service boundaries, standardized UML/DFD specifications, and a modern Java/Node.js tech stack, the platform establishes an enterprise-grade benchmark for digital animal rescue coordination and sustainable shelter management.

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8. GitHub

Repository Link - https://github.com/i4mGHOST/Pawsitive-Connect

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